

OpenClaw Temperature and Humidity Sensor Tutorial

Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [04-AI Voice Interaction Configuration], and [05-Three Dialogue Configuration Tests] in the `Preparation Before Use` directory. If you are only using TUI/WhatsApp, you can skip `04`. This article assumes the basic environment is ready and only covers DHT-related content.

1. Tutorial Objectives

This article demonstrates OpenClaw's natural language control capability through DHT temperature and humidity sensors. After completing this tutorial, readers will be able to:

- Understand the basic approach to reading temperature and humidity with OpenClaw
- Communicate with OpenClaw via WhatsApp, TUI, and voice commands
- Complete four DHT cases: single reading, comfort judgment, threshold alert, OLED display
- Understand the complete chain from "user question" to "sensor reading" and "result feedback"

2. Hardware Preparation

- OpenClaw main control board
- DHT temperature and humidity sensor `[default DHT11]`
- `[Optional]` OLED display screen
- `[Optional]` Microphone module
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial:

- Low-level driver code: `/home/pi/hardware_hal_demo/`
- Temperature and humidity reading command: `/home/pi/hardware_hal_demo/dht`
- Temperature and humidity display with OLED: `/home/pi/hardware_hal_demo/dht_oled`
- Voice entry: `/home/pi/voice_to_openclaw/src/main.py`

The `dht` tool already supports the following capabilities:

- Read current temperature and humidity
- Output results in `json` or `text` format

- Use `mock` fake data for local joint debugging

```
dht_oled #Display temperature and humidity data on OLED
dht #Feedback `json` result
```

If you want to display temperature and humidity results directly on OLED, you can also use `dht_oled`.

Note: If you want to write your own tutorial to read temperature and humidity data from the Python driver library, it is recommended to read DHT data at a two-second interval, as DHT sensors are more suitable for 2s sampling

4. Testing Temperature and Humidity Reading

Before starting this tutorial, please complete an `openclaw tui` basic conversation self-check first; if you haven't done this yet, see [05-Three Dialogue Configuration Tests]. After confirming OpenClaw can respond normally, proceed with the DHT-specific test.

Enter TUI by typing the following in the terminal:

```
openclaw tui
```

```
pi@raspberrypi: ~
File Edit Tabs Help
pi@raspberrypi:~ $ openclaw tui
[11] OpenClaw 2026.3.12 (6472949)
Your terminal just grew claws-type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_u
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)
```

You can try the following in the terminal:

- Read current temperature and humidity
- What is the current environmental humidity
- Help me judge if the current environment is comfortable
- Remind me if the temperature is too high

```
pi@raspberrypi: ~
File Edit Tabs Help
Read current temperature and humidity

Current temperature and humidity readings:

- Temperature: 29.9°C (85.8°F)
- Humidity: 52%

Sensor: DHT11 ✓

What is the current environmental humidity

Current environmental humidity: 52%

(Temperature is 29.9°C if you need that too) 📡
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
48k/1.0m (5%)
```

5. Three Ways to Communicate with OpenClaw

Method	Suitable Scenario	Advantages	Suggested Use
WhatsApp	Remote control, classroom demonstration	Easy to connect, suitable for multi-person experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First to get the control link working
Voice Code	Voice interaction, complete AI experience	Closest to real conversation experience	As the final demonstration solution

If you have the optional AI voice module + speaker, the TUI and WhatsApp conversations need to broadcast OpenClaw's replies. You can run the following program in the terminal. This is not needed for AI voice module conversations.

- Modify the TTS enable parameter

```
/home/pi/voice_to_openclaw/.env
```

```
ENABLE_TTS=true # This parameter controls whether to broadcast replies. After saving, run the program below in the terminal.
```

- Terminal input

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

```
○ (.venv) pi@raspberrypi:~/voice_to_openclaw $ python -m src.openclaw_session_tts  
[系统] OpenClaw 会话播报器已启动  
[系统] 监听目录=/home/pi/.openclaw/agents/main/sessions  
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp Solution

For WhatsApp integration steps, please refer directly to [05-Three Dialogue Configuration Tests]. This section only includes dialogue examples suitable for the DHT tutorial.

After creating, you can converse directly, for example:

- Read current temperature and humidity
- Judge if it's currently hot
- Remind me if the humidity is too low

19:53



< 2



+86 153 3885 7526 (你)

给自己发消息

OLED

19:52 ✓✓

Done! "welcome to openclaw" is now displayed on the OLED screen. The text was split across two lines due to the 128-character width limit of the display.

19:52 ✓✓

Read current temperature and humidity

19:53 ✓✓

Current readings from the DHT11 sensor:

- **Temperature:** 29.7°C (85.5°F)
- **Humidity:** 52%

The environment is fairly warm and moderately humid.

19:53 ✓✓



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space

return



5.2 TUI Solution

If you have already completed the TUI self-check from chapter 0, simply type the following command in the terminal to continue the conversation:

```
openclaw tui
```

```
pi@raspberrypi:~ $ openclaw tui
[Hi] OpenClaw 2026.3.12 (6472949)
Your terminal just grew claws-type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | baillian/qwen3.5-plus | tokens 0/1.0m (0%)
```

5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up flow, please refer directly to [04-AI Voice Interaction Configuration].

You need to say "Hi,yahboom" to wake up for conversation.

After waking up, you can directly speak control content, for example:

- What is the current temperature
- Help me read the humidity
- Tell me if it's too hot and humid

```
pi@raspberrypi: ~/voice_to_openclaw
File Edit Tabs Help
[系统] ASR=xunfei_ws/1dc/en_us | 发布=gateway_chat | TTS=j/1/xunfei_ws | 唤醒=中
口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Read current temperature and humidity.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Current readings from the DHT11 sensor:

- **Temperature:** 30°C
  (86°F)
- **Humidity:** 51%

The sensor is working well! 🌡️💧
[信息] 回复摘要失败, 已退回本地清洗播报
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 6810 ms 语音, 正在转写

[你] It's too hot. Please, let me know.
[发送] 正在交给 OpenClaw
```

6. Comprehensive Cases

Note: All demonstrations below use the `openclaw tui` method for conversation. You can also choose WhatsApp or voice to complete.

6.1 Case 1: Single Reading

Experiment Objective

Get current temperature and humidity once for the most basic environmental sampling.

Effect Description

After OpenClaw receives the command, it will call the DHT driver to read current temperature and humidity and return the results to the user in natural language.

Example Commands

- Read current temperature and humidity
- What is the current indoor temperature
- Help me check the current humidity

6.2 Case 2: Comfort Judgment

Experiment Objective

Have OpenClaw make a simple judgment about the current environment after reading the data.

Effect Description

Besides directly reporting numbers, OpenClaw can also roughly judge "too hot", "too cold", "too dry", "too humid" based on temperature and humidity, making the results closer to natural conversation.

Example Commands

- Help me judge if the current environment is comfortable
- Is the current temperature and humidity suitable for prolonged office work
- Give me a short evaluation based on temperature and humidity

6.3 Case 3: Threshold Alert

Experiment Objective

Have OpenClaw give an alert when temperature or humidity exceeds a set range.

Effect Description

When the temperature is too high, too low, or the humidity is too high or too low, OpenClaw can directly output alert information, and can also link with RGB lights for more obvious feedback.

Example Commands

- Remind me if the temperature exceeds 30 degrees
- If humidity is below 40%, prompt that the air is too dry
- Has the alert condition been reached now

Extensible Play

- Link with RGB for red, yellow, blue three-color alerts
- Link with OLED to display alert text
- Add a buzzer for stronger alerts

6.4 Case 4: OLED Display

Experiment Objective

Display the just-read temperature and humidity results directly on OLED, forming a complete data display chain.

Effect Description

After OpenClaw completes an environmental sampling, it can not only reply the results to the user but also synchronously write temperature and humidity to the OLED screen, forming a complete "reading + display" effect.

Example Commands

- Read current temperature and humidity and display on OLED
- Write temperature and humidity to the screen
- Make a simple temperature and humidity display page

Extensible Play

- Periodically refresh OLED content
- Make a temperature and humidity dashboard
- Link with the weather station project to form a complete application